

Novel drug targets and molecular mechanisms for sarcopenia based on systems biology

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ABSTRACT

Sarcopenia is a major public health concern among older adults, leading to disabilities, falls, fractures, and mortality. This study aimed to elucidate the pathophysiological mechanisms of sarcopenia and identify potential therapeutic targets using systems biology approaches. RNA-seq data from muscle biopsies of 24 sarcopenic and 29 healthy individuals from a previous cohort were analysed. Differential expression, gene set enrichment, gene co-expression network, and topology analyses were conducted to identify target genes implicated in sarcopenia pathogenesis, resulting in the selection of 6 hub genes (PDHX, AGL, SEMA6C, CASQ1, MYORG, and CCDC69). A drug repurposing approach was then employed to identify new pharmacological treatment options for sarcopenia (clofibrate, troglitazone, withaferin-a, palbociclib, MG-132, bortezomib). Finally, validation experiments in muscle cell line (C2C12) revealed MG-132 and troglitazone as promising candidates for sarcopenia treatment. Our approach, based on systems biology and drug repositioning, provides insight into the molecular mechanisms of sarcopenia and offers potential new treatment options using existing drugs.

1. Introduction

Sarcopenia is defined as the loss of skeletal muscle and the decrease of its function by ageing [1]. It is one of the important health issues in the old population, causing fractures, prolonged hospitalization, functional decline, and even mortality [2]. The prevalence of the disease among individuals aged 60–70 ranges from 5 % to 13 %, whereas it reaches nearly 50 % among those aged 80 years and older [3]. While there is no approved drug for the treatment of sarcopenia, current management options focus on nutrition, exercise, hormone replacement therapies, and biotherapies [1,4]. Therefore, the identification of novel drug targets may present possibilities for the development of innovative therapeutic interventions.

Systems biology is an emerging field that seeks to understand biological processes at a deeper level by integrating data from multiple omics technologies. One of the key applications of systems biology is in mechanism-based drug discovery, where computational models are created for specific regulatory cascades in human cells [5]. Network

models generated from omics data can provide insights into disease mechanisms and enable the identification of target genes through the analysis of intramodular topology [6]. Previous studies utilizing this approach have proposed novel treatment alternatives for diverse illnesses, demonstrating the potential for systems biology to drive the development of new therapeutics [7–9]. By integrating data from various omics technologies, researchers can identify molecular signatures that are associated with specific disease states, allowing for earlier diagnosis and a more accurate prognosis [10]. Systems biology approaches have already been applied to a wide range of diseases, including cancer, Alzheimer's disease, and cardiovascular disease, with promising results [11–17]. The use of systems biology in biomarker discovery and validation could greatly improve patient outcomes and lead to more personalized treatments.

Drug repurposing is an important strategy that can accelerate the development of new therapies for a variety of diseases, including sarcopenia. By identifying alternative medical applications for approved drugs, drug repurposing can save time and money compared to

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developing new drugs from scratch [18]. Although some drugs have been repurposed for sarcopenia treatment, such as metformin, there is still a lack of FDA approval for their use in this context [19]. The integration of systems biology approaches in drug repurposing has the potential to identify novel drug targets and provide insights into the underlying mechanisms of diseases. For example, network-based approaches can be used to identify new drug targets by analysing protein-protein interaction networks and predicting the effects of drug-target interactions [17]. By combining network analysis with experimental validation, this approach has been successful in identifying new drug targets for a range of diseases, including cancer and cardiovascular [14,20]. Moreover, systems biology can aid in the identification of patient subgroups that may benefit from repurposed drugs, enabling more personalized treatments [11,15].

Our research aimed to explore the molecular mechanisms that cause sarcopenia through systemic analysis and to pinpoint potential drug targets that can be utilized to develop effective treatment options for this condition. In addition, we were able to detect drugs that had been repurposed and demonstrated their effectiveness through wet-lab experiments. Fig. 1 illustrates the procedures that were followed for this study.

2. Materials and methods

2.1. Data and preparation

The present study utilized data from a previous study on sarcopenia [21]. The cohort has 93 samples, including 19 young healthy, 29 old healthy, 24 sarcopenic and 4 unclassified bulk RNA-seq and 17 single-nuclei RNA-seq. Of those 29 old healthy and 24 sarcopenic samples were used in our study for analysis to focus on differences between sarcopenic and healthy aged populations. In the previous study, Illumina HiSeq 4000 sequencing technology was employed to obtain sequence data and the STAR aligner was utilized to generate raw counts, with the GRCh38 reference genome being used [21]. Notably, genes with a total count of less than 10 were excluded from the analysis. Gene count profiles of the cohort were downloaded in CSV format from the Gene Expression Omnibus website (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE167186>, Access Date: 10 October 2022). These data were subsequently analysed to investigate potential biomarkers of sarcopenia.

2.2. Differential expression analysis and gene ontology analysis

DESeq2 package was used in R Studio to perform Differential Expression Analysis [22]. First, a matrix was constructed between metadata and raw count values of sarcopenic and old healthy groups. Then, the analysis was executed with old healthy selected as the

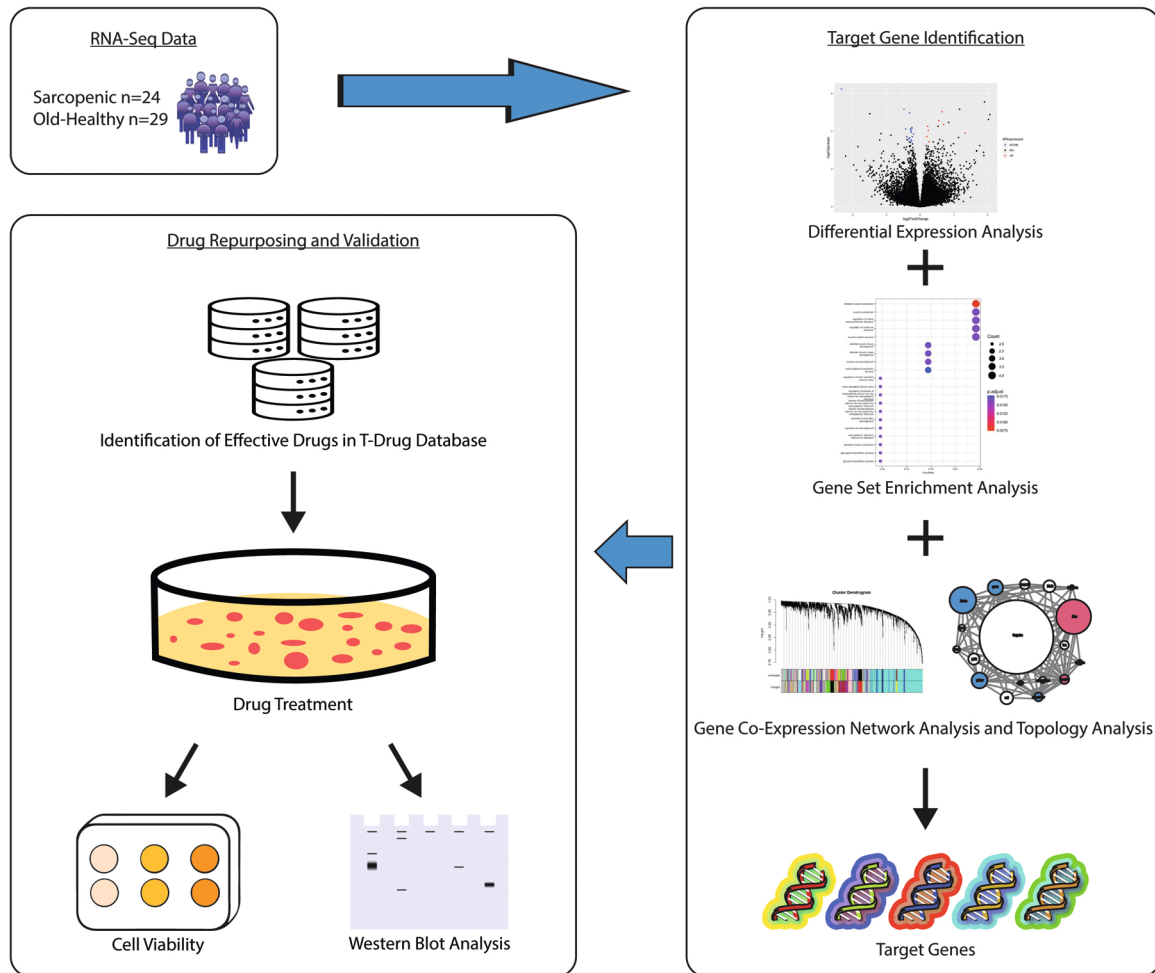


Fig. 1. A diagram outlining a structured strategy of drug repurposing for the treatment of sarcopenia. We used RNA sequencing data and gene counts from a previous cohort to identify target genes. To do so, we performed differential expression analysis, gene set enrichment analysis, gene co-expression network analysis, and topology analysis. We then used a drug repurposing strategy to activate these genes and validated our findings with in vitro experiments.

reference condition to identify differentially expressed genes (DEGs). The Benjamini-Hochberg (BH) approach was used to adjust the false discovery rate (FDR). To find significant DEGs, FDR 0.05 was applied.

Gene ontology analysis (GOA) was performed using the "enrichGo" function of the clusterProfiler package in R, which determines the list of highly enriched genes in biological process pathways by using the hypergeometric distribution [23]. To find substantially enriched pathways, $p < 0.05$ was employed as the cut-off value. The clusterProfiler sub-package, enrichplot, was operated to visualize GOA results.

2.3. Weighted gene co-expression network analysis

The weighted gene co-expression network analysis (GCN) technique was utilized to establish a gene co-expression network using the R package "WGCNA" [24]. Metadata and raw count values of the sarcopenia cohort were used as input. Initially, we conducted a quality assurance process by utilizing the "goodSamplesGenes" function to eliminate genes with many incomplete records. Then, we created a clustering tree map to identify and eliminate outlier samples. Later, a matrix was created from the input data, and genes which count less than 15 in along with 75 % of the total samples were removed for downstream analysis. The DESeq2 package was used for variance stabilization and to get normalised counts.

The correlation coefficient was computed between every pair of genes to generate a gene expression similarity matrix, which was then transformed into an adjacency matrix. The "pickSoftThreshold" function was utilized to determine the ideal soft thresholding value, resulting in a scale-free network with a scale independence of 0.8. Then, the adjacency matrix was converted to a topological overlap matrix to display the relationships between genes and hierarchical clustering. Dynamic cut tree methods were employed to identify co-expression gene modules with a minimum connectivity score cut-off of 0.5. These modules that had a positive correlation with the disease state of sarcopenia were termed upregulated modules, and negatively correlated ones were named downregulated modules. In the final step, intramodular analysis was used to identify driver genes by calculating the module membership and the associated p-values.

2.4. Identification of target genes

To identify the most influential genes in the upregulated and downregulated modules that were previously identified, a topology analysis was conducted to prioritize hub genes with extensive impacts on the network of modules. All implicated genes' module membership parameters were calculated, and then the topological score was used to arrange all genes into GCN modules in descending order. Afterwards, the top 20 % of genes of each module were extracted.

First, we looked intersection between upregulated-downregulated DEGs and %20 of upregulated-downregulated modules, respectively. In the second step, we inspected these genes deeper in the Human Protein Atlas Database [25]. Genes which have specific RNA expression levels in muscle tissue were selected as our targets.

2.5. Drug repositioning

In our previous study, we proposed a drug repositioning method based on the similarity analysis of transcriptomics signature profiles of human cells perturbed by drug treatment and genetic treatment [16]. The hypothesis of this method is to identify a drug that causes a significant change in gene expression levels in cells that mimics the effect of inhibiting or activating the specific target genes. Based on this method, we constructed T-Drug (<http://drug-reposition.com/>) which is a web tool that helps to identify drugs specifically targeting a disease's therapeutic gene [16,17]. In this web tool, users can query a gene and make predictions about the potentially useful drugs that can inhibit or activate the protein expression of these queried genes.

Drug-perturbation matrix was downloaded for each target gene for the desired effect type in all perturbation data types (shRNA, CRISPR-CAS9, and plasmid overexpression) from the T-drug website with a 0.4 correlation coefficient cut-off. Later, the drugs were sorted in descending order by the number of hit cells criterion, showing how many cell lines where the drug-target association exceeds the correlation coefficient score. Then, the top 200 drugs of each gene were extracted for the following steps. Ultimately, the relationship between each drug's primary target and our targets was inspected by using the iNetModels database developed by our research group [26,27]. Drugs that have a preferred association and can cause upregulation in the expression of the target genes were selected as our candidates.

2.6. Cell culture and induction of sarcopenia cell model

The C2C12 myoblast line (91031101, Sigma Aldrich, USA) was used for the sarcopenia cell model. The myoblasts were maintained in modified Eagle's medium (DMEM, D-6429 Sigma Aldrich) supplemented with 10 % fetal bovine serum (FBS, F7524, Sigma-Aldrich), 1 % Penicillin/Streptomycin (P/S, P4333, Sigma Aldrich) in a humidified incubator with 5 % CO₂ at 37°C. N-acetyl-D-sphingosine (C2-ceramide, A7191, Sigma Aldrich) was used for inducing sarcopenia in the C2C12 cell line. After one day of seeding 3×10^5 cells for each well in a 6-well plate, 40 μ M C2-ceramide and the drugs dissolved in 0.1 % Dimethyl sulfoxide (DMSO, 41639, Sigma-Aldrich) were administered in the maintenance condition for two days. Then, the medium was changed twice, at two-day intervals, with a differentiation medium (DMEM with 2 % FBS and 1 % P/S) to differentiate myoblasts into myotubes. Because ceramide may affect mitochondrial activity in muscle cells, another cell model was created without introducing it [28]. After completion of myotube differentiation, cells were harvested for western blot analysis.

2.7. Cell viability assay

The effects of the drugs on the C2C12 cell viability were tested by MTT (Thiazolyl Blue Tetrazolium Bromide) method. 1×10^4 cells were seeded to per well in a 96-well plate in 100 μ L growth media. After one day of cell seeding, different concentrations of the drugs dissolved in 0.1 % DMSO and 100 μ L of the media were treated on the cells for one day. After the treatment period, 5 mg/ml MTT (M2128, Sigma-Aldrich) solution in PBS (10 μ L) was added to each well. After one hour, the MTT solution and all media were removed from the wells. 100 μ L of DMSO was added and mixed to dissolve the formazan crystals. Cell viability was analysed by measuring the absorbance of the dissolved formazan at a wavelength of 570 nm with a microplate reader (Hidex Sense Beta Plus).

2.8. Western blot

To test the effects of the drugs on the expression of target genes and myogenic markers in the sarcopenia cell model, the cells were washed with phosphate-buffered saline (PBS) solution and then lysed with Cel-Lytic M (C2978, Sigma-Aldrich) lysis buffer containing protease inhibitors (11836170001, Roche). The cell lysates were centrifuged at 12,000 rpm for 10 min and supernatants were collected. Protein Assay Dye Reagent (5000006, Bio-Rad) was used for the determination of protein concentration. The absorbance of the proteins was measured spectrophotometrically at 595 nm by a microplate reader (Hidex Sense Beta Plus). Protein electrophoresis was performed using Mini-PROTEAN® TGXTM Precast Gels (4561086, Bio-Rad) and then the separated proteins were transferred to a Trans-Blot Turbo Mini 0.2 μ m PVDF Transfer Packs membrane (1704158, Bio-Rad) by using Trans-Blot® TurboTM Transfer System (Bio-Rad). The membrane was blocked with 5 % skim milk for 30 min at 4°C with gentle rocking. After blocking, the membrane was treated with primary antibodies: anti-AGL (1:10000 dilution; ab133720, Abcam, UK) anti-PDHx (1:5000 dilution;

HPA038484, Human Protein Atlas, Sweden), anti-Myogenin (1:10000 dilution; ab124800, Abcam), anti-Myosin/MyH3 (1:10000 dilution; ab124205, Abcam), anti-Beta actin (1:20000 dilution; ab8227, Abcam) and anti-Alpha tubulin (1:10000 dilution; ab7291, Abcam) as a loading control overnight at 4°C on a rocking platform. After the treatment of primary antibodies, the membrane was washed three times with TBS-T buffer (A09-7500-100, Medicago). Then, Goat anti-Rabbit IgG-HRP (1:10000 dilution; ab205718, Abcam) and Goat anti-Mouse IgG-HRP (1:10000 dilution; ab6789, Abcam) were treated as the secondary antibody for 30 min at 4°C with gentle rocking. The protein bands on the membrane were revealed using enhanced chemiluminescence substrate (WBLUF0500, Merck) and detected with ImageQuant™ LAS 500 (GE Healthcare).

3. Results

3.1. Different patterns of gene activity and specific functions in sarcopenia

We conducted a global gene expression comparison study between sarcopenic and healthy individuals to identify differences using differential expression analysis. Along 36,694 genes, 1063 of them have a $p < 0.05$ and only 26 of them have a p -adjusted < 0.05 . We continue our analysis with 26 DEGs whereas p -adjusted methods give more robust results in multiple testing [29] Eight of the DEGs were upregulated and 18 were down-regulated in the state of sarcopenia (Fig. 2A). Furthermore, we analysed the functional characteristics of DEGs with increased or decreased expression levels using gene ontology enrichment analysis. No common biological process pattern was found in 8 upregulated genes. However, downregulated DEGs enriched in muscle contraction, store-operated calcium entry, regulation of cation transmembrane transport, muscle cell development, and other metabolic processes such as glycogen, glucan, and carbohydrate synthesis (Fig. 2B).

3.2. Co-expression network analysis highlights the key modules in sarcopenia

We created co-expression networks based on the count data from the cohort study to examine functional gene modules. In the first quality control step, 15,279 genes were removed due to too many missing entries. Then, the hierarchical clustering method identified 2 (X_41, X_78)

outlier samples and excluded them from the downstream analysis (Fig. 3A). After normalising the counts with the Deseq2 package, 3 was selected as soft threshold power due to giving high R^2 results and lower mean connectivity (Fig. 3B). Next, we created an adjacency matrix and employed the topological overlap measure (TOM) and the dynamic tree cut to form clusters. 18 modules were identified that module sizes ranging between 51 and 1739. All modules consist solely of genes with consensus eigengene connectivity of at least 0.5 (Fig. 3C).

For the following step, we measured the Pearson correlation score between modules and the samples' disease state of sarcopenia. Out of all the modules tested, 6 exhibited significant outcomes ($p < 0.05$). Along these modules, 2 of them were identified as upregulated (blue and purple); and 4 were classified as downregulated (green, green-yellow, brown, and yellow) (Fig. 3D). Also, we carried out gene ontology analysis on the gene modules to determine their related biological pathways. Interestingly, upregulated modules were mainly enriched in mitochondrial energy metabolism. Blue module's gene ontology analysis (GOA) results primarily include aerobic respiration, oxidative phosphorylation, and ATP biosynthetic process. Likewise, mitochondrial components were enriched in the purple module such as mitochondrial cytochrome c oxidase assembly, respiratory chain complex IV assembly, and cytochrome complex assembly. Upon examining the downregulated ones, we observed that they exhibited unique patterns that were different from one another. The green module was enriched in skeletal muscle fibre development, skeletal muscle tissue development and transcription process. Brown module involved with muscle contraction, purine metabolism, mitochondrial respiratory chain subunits and ATP metabolism. The yellow module covered the protein catabolic process and vesicle organization. In despite of including 106 different genes, the green-yellow module didn't cluster any specific function in GOA (Fig. 4). Lastly, we calculated the module membership score to identify hub nodes using Pearson correlation of the gene expression levels with module eigengenes.

3.3. Determining target genes

Examining the structure of a network can reveal the significance of a specific gene within it. Hub genes, which have a high score in topology analysis, are critical components of the network and are believed to be good candidates for further investigation as potential targets. To achieve

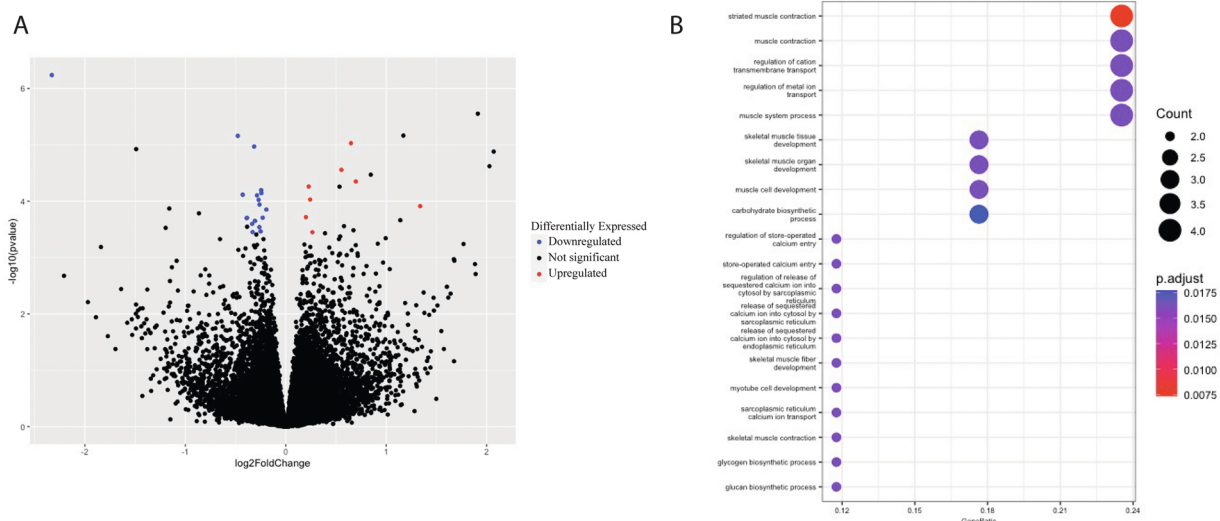


Fig. 2. A) Distribution of differentially expressed genes. Genes that have significant ($FDR < 0.05$) upregulated expressions are shown in red colour; while those with significant ($FDR < 0.05$) downregulated expressions are shown in blue. Genes with FDR scores above 0.05 are displayed in black. B) Top 20 enriched pathways of downregulated DEGs, based on the p -adjusted score. The colour red indicates a smaller p -adjusted value, while blue indicates a higher value. The dots' size represents the number of genes in a specific pathway. Among the downregulated differentially expressed genes (DEGs), the metabolic process of striated muscle contractions is the most frequently expressed.

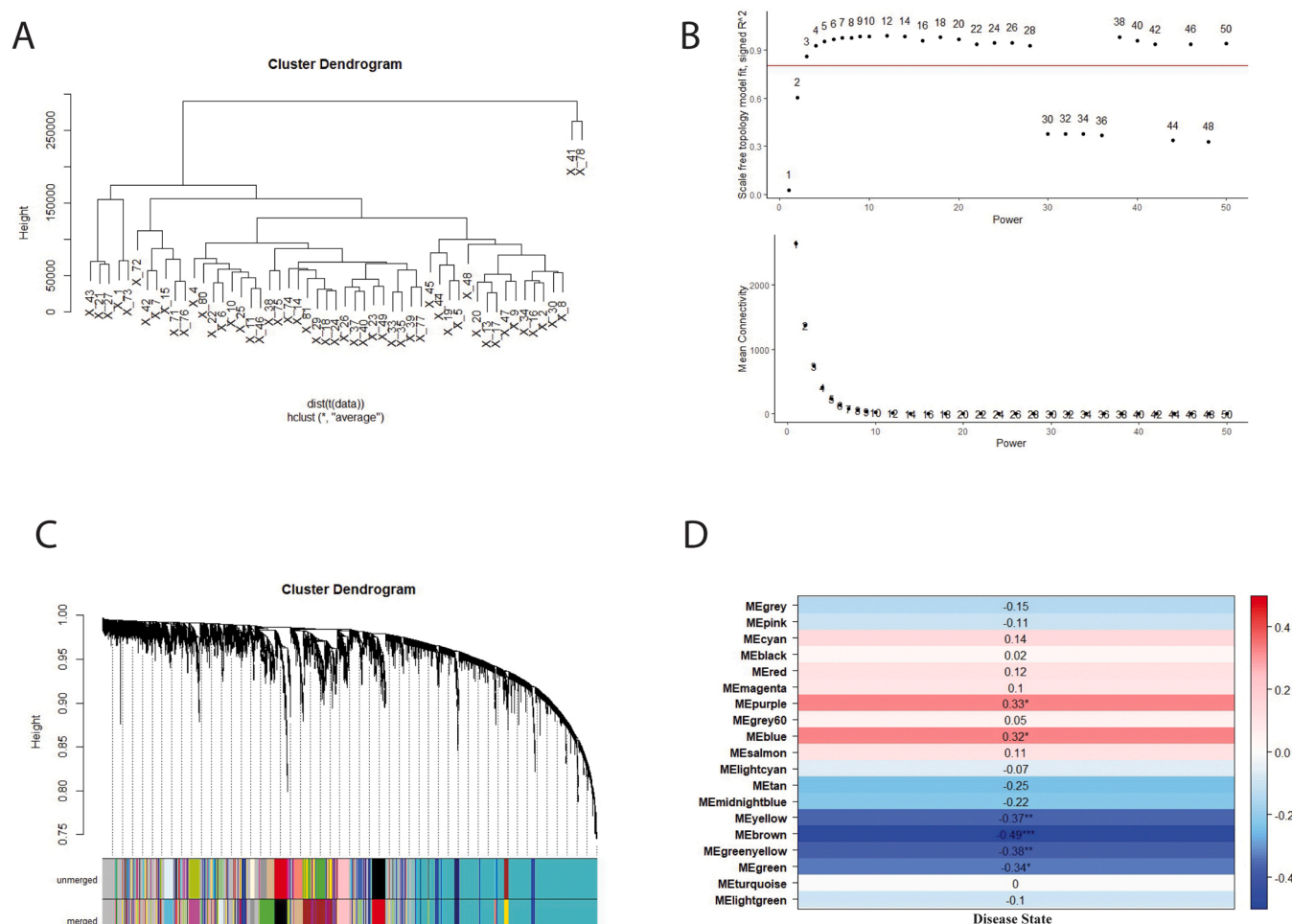


Fig. 3. A) The dendrogram illustrating the hierarchical clustering of the samples from the sarcopenia cohort was generated. However, two samples (X_41 and X_78) were excluded from the analysis as they were considered outliers. B) Graphs depicting the mean connectivity and fit of the scale-free topology model were produced. The line corresponding to the y-intercept was set at 0.8. Based on the results showing a high R² value and reduced mean connectivity, the soft threshold power of 3 was chosen. C) Profiles of cluster dendrograms and module detection were displayed. The modules' colours were presented before and after merging below the dendrogram. D) The heatmap presented shows the Pearson correlation between the module eigengenes and sarcopenia disease state. A positive correlation is depicted in red, while a negative correlation is depicted in blue. The level of significance is denoted by asterisks, with a threshold of $p < 0.05$. Any genes that were not included in any other module were categorized into the grey module.

this, we conducted a network topology analysis by assessing the module membership of genes in both upregulated and downregulated modules. Then, we arranged all the genes in each upregulated and downregulated module in descending order based on their module membership parameters and selected the top 20 % of genes from each module. As a result, 96 genes from blue and 22 genes from purple were extracted. Subsequently, we discovered one gene (ZFHX3) common in upregulated modules and upregulated DEGs. Likewise, 65 genes were taken from the brown module, 36 genes from the green module, 20 genes from the green-yellow module, and 39 genes from the yellow module. Afterwards, we found eight genes (APPL1, PDHX, SEMA6C, AGL, CASQ1, SLC37A4, MYORG, and CCDC69) from the intersection between the downregulated modules and downregulated DEGs (Fig. 5).

Furthermore, we investigated the genes' tissue RNA expression profiles in The Human Proteome Atlas. [25] Our upregulated target candidate, ZFHX3, is mainly enriched in brain tissue. Inside down-regulated candidates, APPL1 showed low tissue specificity and SLC37A4 is mainly enriched in the liver. However, the rest of them (PDHX, SEMA6C, AGL, CASQ1, MYORG, and CCDC69) are mainly expressed in muscle tissue. Therefore, our final decision in terms of detecting our targets includes the 6 genes stated above.

3.4. Drug repositioning

We carried out a drug repositioning approach and found effective medications that can regulate the target genes by combining the sarcopenia transcriptomic signature profiles linked to gene activation, over-expression and chemical alteration from the T-drug database (<https://drug-reposition.com/>). We conducted our repositioning analysis for PDHX and AGL because data for 2 out of 6 target genes were solely accessible in the database.

We obtained the transcriptomic signature profiles related to activating PDHX in 13 various cell lines, and the over-expression of AGL in 9 diverse cell lines. Then, gene activation and drug treatment similarity matrix for target genes were downloaded from the database. 6235 drugs and 178 drugs were found to activate PDHX and AGL, respectively. Next, we interrogated the connection between each drug's primary target and our target genes individually by using the iNetModels website.

We saw that Clofibrate (PPAR receptor agonist) targets PPARA which has a positive relationship with PDHX. Another drug named Troglitazone (PPAR receptor agonist) has a similar target, PPARG has a positive correlation with PDHX as well. One of the Withaferin-a (IKK inhibitor) targets, NAMPT shows negative ties with AGL. A CDK inhibitor, Palbociclib targets CDK4 that has negative correlations with AGL.

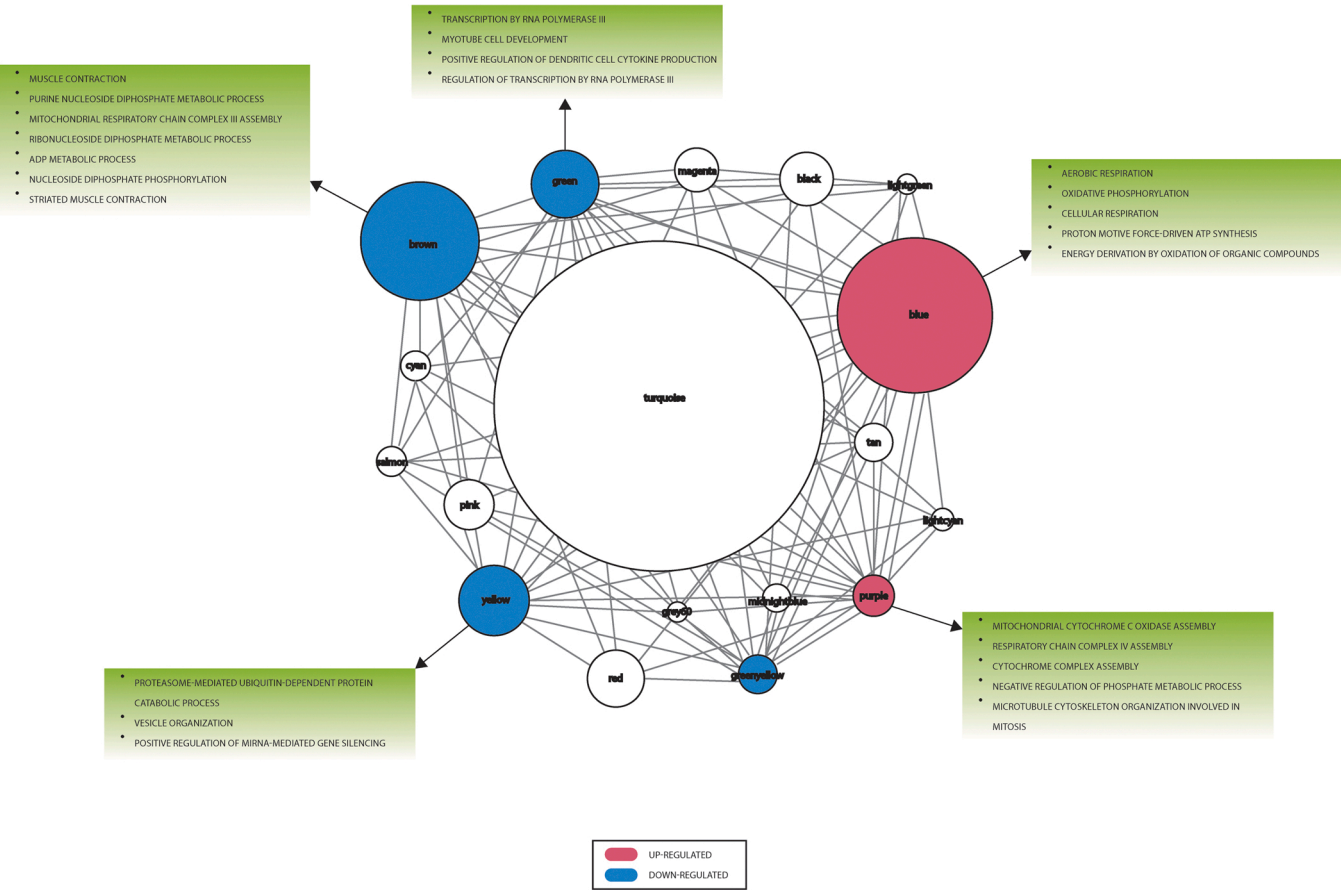


Fig. 4. Functional modules and related biological processes in sarcopenia cohort. Network was built with a minimum connectivity score of 0.5 between eigengenes in each module. Red colour shows upregulation and blue colour shows downregulation in the disease state. Circle diameters represent the size of modules.

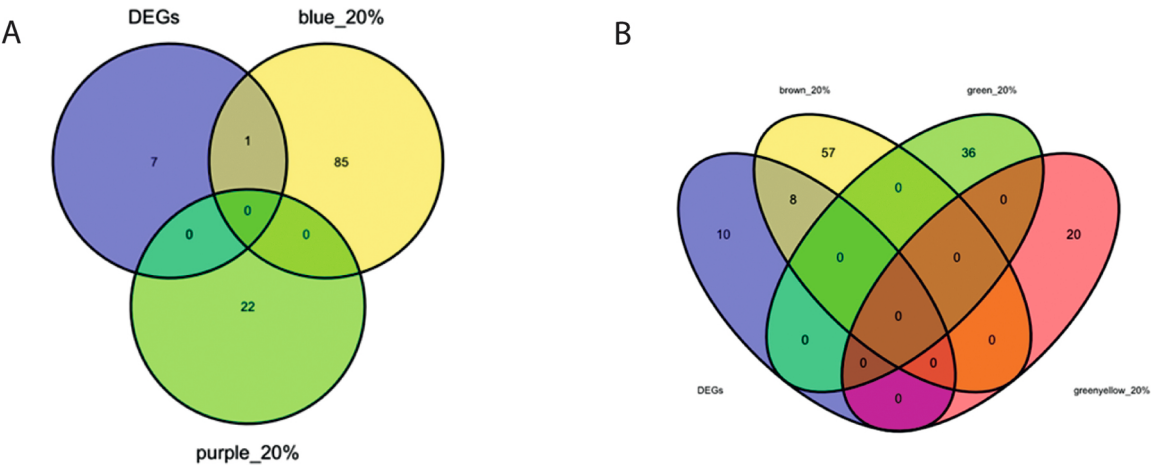


Fig. 5. The intersection between the top 20 % genes of upregulated modules/upregulated differentially expressed genes (DEGs) (left) and the top 20 % genes of downregulated modules/downregulated DEGs (right). Two genes were discovered to be common between the upregulated modules and upregulated DEGs, whereas eight genes were found to be common between the downregulated modules and downregulated DEGs.

Target of MG-132 (protease inhibitor), PSMB1 has negative affairs with AGL. Two of Bortezomib's (NFKB pathway inhibitor) targets, PSMB1 and PSMB2 display negative relations with AGL. In total, 6 drugs were found to activate our target genes and can be used in the treatment of sarcopenia based on our in-silico analyses (Fig. 6).

3.5. Validation of the efficacy of drugs in the sarcopenia cell model

The C2C12 myoblast cell line was chosen for sarcopenia research due to its ability to differentiate into myotubes, which are multinucleated cells that closely resemble skeletal muscle fibres. It has been used in many previous studies due to its well-characterized nature, responsiveness to relevant stimuli, suitability for genetic manipulation, and consistent performance in vitro experimentation [30–37]. Before testing

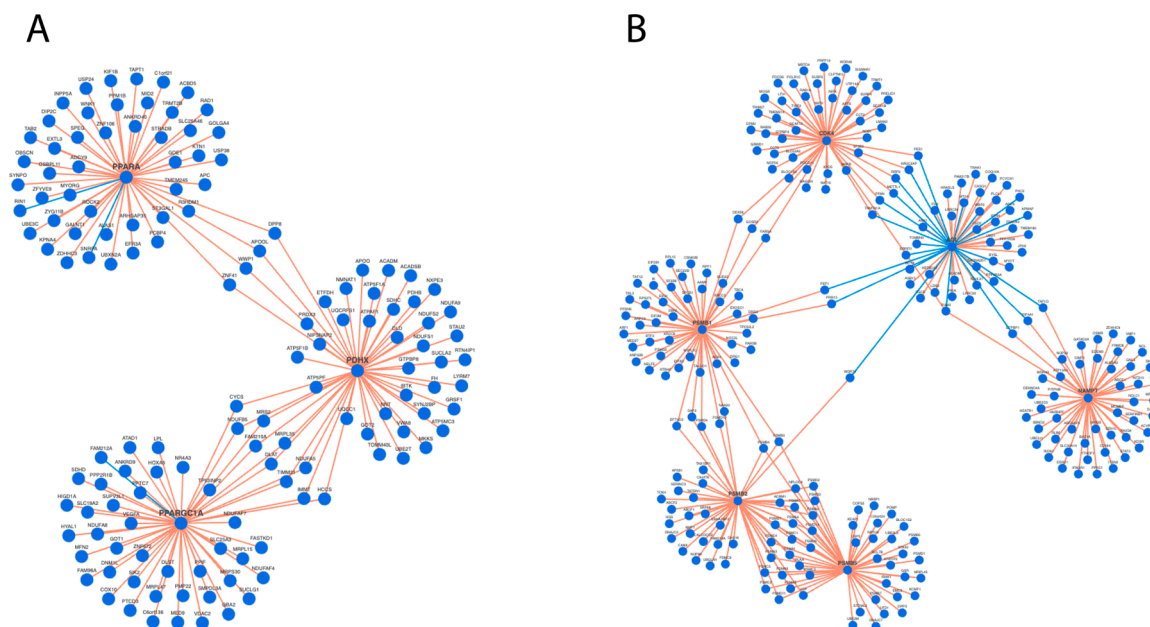


Fig. 6. A) The connection between particular drug targets and PDHX was examined, and it is expected that the expression of PDHX will increase when PPARA and PPARG are activated by drugs, as there is a robust positive correlation between them. B) The connection between specific drug targets and AGL was investigated by inetModels2 (<https://inetmodels.com/>), and it was predicted that inhibiting NAMPT, PSMB1, PSMB2, and PSMB5 by drugs would increase the expression level of AGL since they exhibit a negative correlation. Positive correlations are indicated by red lines, while negative correlations are indicated by blue lines.

the efficacy of the drugs in the sarcopenia cell model, the effects of different concentrations of each drug on the viability of C2C12 cells were determined and non-toxic optimal doses were selected as follows: Palbociclib (5 μ M), MG-132 (25 nM), Bortezomib (1 nM), Withaferin-a (50 nM), Clofibric-acid (5 μ M) and Troglitazone (0.5 μ M). Then, the selected doses of each drug were administered in the sarcopenia cell model to test the effects of the drugs on the expression of target genes (AGL and PDHX) and myogenic markers (myogenin and myosin heavy chain) for the therapy of sarcopenia. It was observed that the protein expression levels of AGL and PDHX in sarcopenia induced by C2-ceramide were decreased compared to the control. In parallel with the decrease in these target genes, myogenin and myosin heavy chain (MyH3) levels were reduced. Therefore, the down-regulated target genes were confirmed in the sarcopenia *in vitro*.

Furthermore, it was determined that MG-132 and troglitazone ameliorated the expression of AGL and PDHX, respectively. Also, it was demonstrated that myogenic markers were increased compared to sarcopenia-induced cells depending on the enhancement in the expression of target proteins. Moreover, it was noted that all the repurposed drugs, except for Palbociclib, resulted in elevated levels of myogenin and myosin heavy chain expression. We also observed the myogenesis facilitation ability of repurposed compounds. Compounds were treated for 2 days without ceramide into C2C12 cells. After an additional 4 days of myogenesis, we tested MyH3 and myogenin protein expression levels with western blot analysis. All repurposed compounds showed highly increased MyH3 expression; Palbociclib (143 %), MG-132 (125 %), Bortezomib (136 %), Withaferin-a (143 %), Clofibric-acid (142 %) and Troglitazone (142 %). Except Palbociclib (90 %), all other repurposed compounds increased myogenin expression; MG-132 (152 %), Bortezomib (157 %), Withaferin-a (159 %), Clofibric-acid (153 %) and Troglitazone (154 %). Ultimately, the efficacy of the target genes (AGL and PDHX) and the effectiveness of MG-132 and troglitazone on the expression of the target genes were validated in the sarcopenia cell model *in vitro* (Fig. 7).

4. Discussion

Sarcopenia is defined as progressive muscle and function loss in the

ageing population. It is an important public health problem which is causing falls, fractures, disabilities and death. Further efforts are required to identify new drug targets and medications to find better treatment options for the disease. Systems biology is a developing area for mechanism-based drug discovery computational models and functional analyses of multi-omics data. This research discovered targets and effective drugs for curing sarcopenia by integrating GCN, DEG, GO, and gene perturbation expression profiles from various data sources. Consequently, we identified PDHX and AGL as target genes which are amenable to drug intervention. They play essential roles in energy production and glycogen metabolism. In addition, we suggested MG-132 and troglitazone for the treatment based on their validated results in our sarcopenia cell model experiments.

Our study contributes to the growing body of literature on sarcopenia by providing new insights into the molecular mechanisms involved in the disease. For instance, PDHX gene encodes pyruvate dehydrogenase component X protein, which is necessary to sustain the functioning of pyruvate dehydrogenase and generate ATP [38]. Pyruvate dehydrogenase is a multi-enzyme complex situated in the matrix of mitochondria that catalyses the conversion of pyruvate to acetyl-coenzyme A. This is the initial irreversible step in the oxidation of carbohydrates [39]. Previous research showed a strong link between impaired mitochondrial function and sarcopenia [40]. Besides, deletion in the PDHX gene results in pyruvate dehydrogenase deficiency, lactic acidosis, and different levels of neurological disorders in clinical presentation [41]. Similarly, the AGL gene produces the glycogen-debranching enzyme, which is crucial in the process of breaking down glycogen [42]. It is a well-studied gene because its absence results in Glycogen Storage Disease Type III (GSD III) [43]. GSD III has 4 different subtypes, however, most individuals with the disease are deficient in the glycogen-debranching enzyme in both the liver and muscle (type IIIa) [44]. Clinical presentation of both GSD III and sarcopenia has similarities, such as muscle atrophy, weakness and peripheral neuropathy [45]. While no previous research shows the link between AGL gene down-regulation and sarcopenia, our study results suggest that defects in glycogen metabolism can be important in the pathophysiology of the ailment as well.

The present study provides further evidence supporting the potential

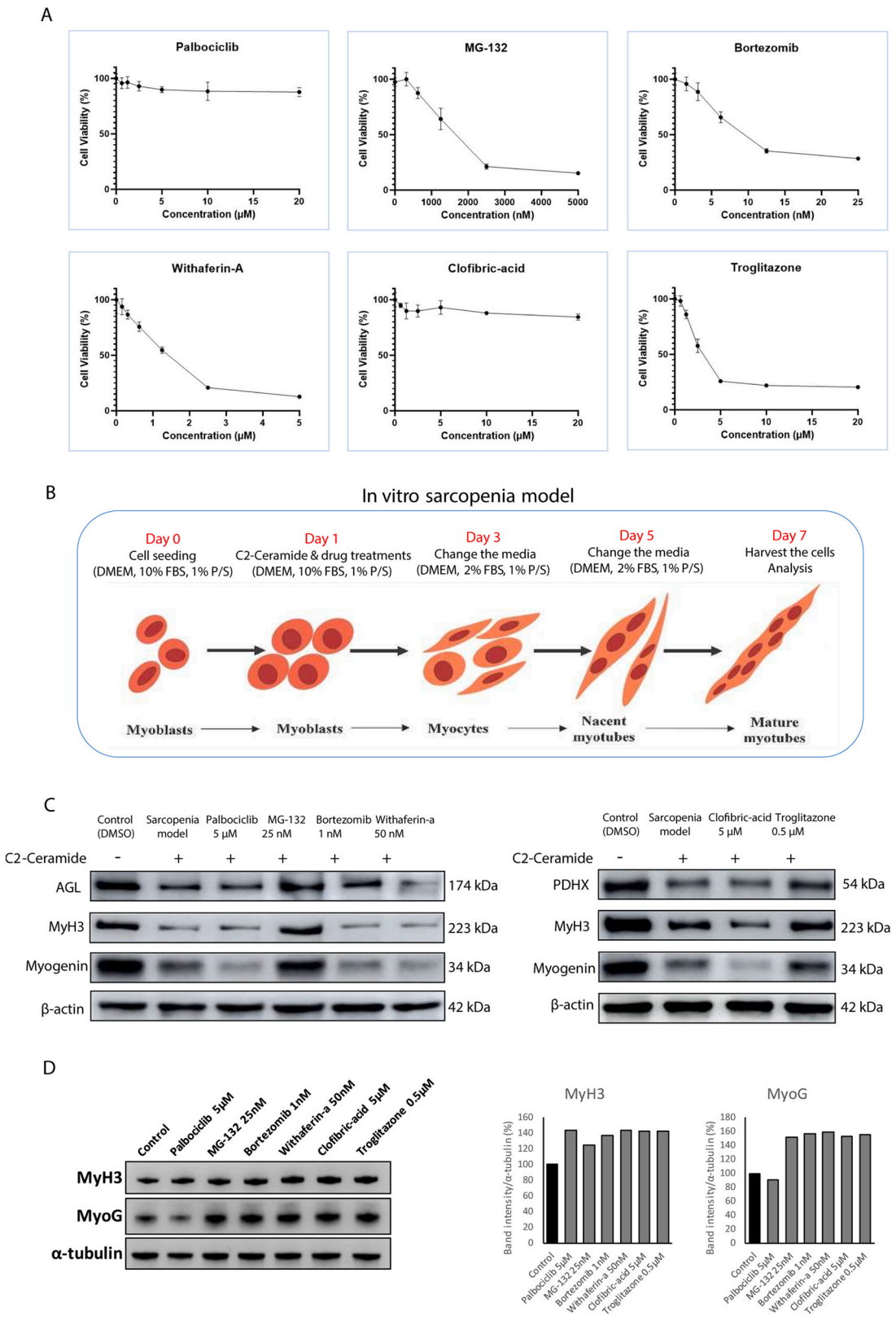


Fig. 7. A) The efficacy of the target genes (AGL and PDHX) and the effect of the drugs on the activation of the target genes in the sarcopenia cell model. A. The toxic effect of each drug on cell viability was assessed by MTT. Each measurement was performed in triplicate. B) Induction of sarcopenia cell model via C2-Ceramide in C2C12 cell line and the stages of differentiation of myoblast cell to myotubes. C) The activation of down-regulated target genes (AGL and PDHX) via drugs with optimized doses was illustrated by Western blot. Myogenin and myosin heavy chain (MyH3) were used to indicate myogenic markers, and beta-actin was used as a loading control. D) Western blotting was used to show myogenic marker expressions in cell models not treated with ceramide, using optimized drug doses. α-tubulin levels served as a reference protein. MyH3 represents myosin heavy chain, and MyoG stands for Myogenin.

of MG-132 and Troglitazone as drug candidates for muscle atrophy and cardiomyopathy, respectively. Our findings are compatible with previous literature and suggest that these drugs may be effective therapeutic options for these conditions. MG-132 is a proteasome inhibitor that suppresses the ubiquitin-proteasome pathway [46]. This pathway plays a significant role in muscle atrophy, as skeletal muscles undergo atrophy under various conditions, and myofibrillar proteins are degraded through the ubiquitin-proteasome system [47]. Recent research has provided evidence supporting the effectiveness of MG-132 in mitigating muscle atrophy[48]. Sakai et al. (2020) observed increased expression levels of ubiquitinated proteins in atrophic muscle cells and their downregulation after introducing MG-132 in mice [48]. This study highlights the potential for MG-132 to serve as a therapeutic option for muscle atrophy.

Troglitazone belongs to the family of oral antidiabetic drugs and is known for its ability to reduce insulin resistance and serum triglyceride levels [49]. A rodent study conducted by Unger and Orci (2002) has also highlighted its potential to improve myocardial contractility in cases of cardiomyopathy [50]. They found that troglitazone administration resulted in increased myocardial contractility in cardiomyopathic rats. Although our results have similarities to previous research, it's important to note that previous ones were not specifically conducted on the elderly sarcopenic group. They focused on the success of drugs in treating muscle atrophy, which was developed as a secondary condition alongside another disease.

The study has some limitations, including a small sample size of 53 publicly available sarcopenia samples. Future research could expand this through larger cohort studies collecting bulk RNA samples from patients. Also, we only used cell models for validation due to budget constraints. Nevertheless, our ultimate aim is to validate our findings using animal models and eventually test the treatment's effectiveness in patients through future clinical trials.

5. Conclusion

In summary, we employed system biology approach to identify target genes for sarcopenia (PDHX, AGL, SEMA6C, CASQ1, MYORG, and CCDC69) and shed light on underlying mechanisms such as reduced energy production and glycogen metabolism. Through drug repurposing and *in vitro* experiments, we suggested the use of MG-132 and troglitazone as potential therapeutic agents for treatment of sarcopenia.

Ethics Approval and Consent to Participate

Not applicable.

Authors' contributions

AM conceived the study, managed the project, and acquired funding. AC organized the data and conducted computational analyses. AC, XL, and CZ aided in the development of computational methodology. MO and WK carried out laboratory experiments. AM, OA, and CZ supervised the study. AC drafted and revised the manuscript. All authors reviewed and accepted the final version of the manuscript.

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CRedit authorship contribution statement

Woonghee Kim: Formal analysis. Mehmet Ozcan: Visualization, Formal analysis. Atakan Ceyhan: Writing – review & editing, Writing –

Table 1

List of repurposed drugs with their target genes and intended effect type.

Target	Effect Type	Drugs
PDHX	Activate	clofibric-acid troglitazone
AGL	Activate	withaferin-a palbociclib MG-132 bortezomib

PDHX: Pyruvate Dehydrogenase Complex Component X

AGL: Amylo-alpha-1, 6-glucosidase, 4-alpha-glucanotransferase

original draft, Visualization, Investigation, Formal analysis, Data curation, Conceptualization. **Adil Mardinoglu:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Cheng Zhang:** Supervision. **Ozlem Altay:** Writing – review & editing, Supervision. **Xiangyu Li:** Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Data availability

Data will be made available on request.

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Consent for Publication

Not applicable.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.biopha.2024.116920](https://doi.org/10.1016/j.biopha.2024.116920).

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